

CONTROLLED LESSON

Singular Value Decomposition, PCA, and Dimensionality Reduction

Preserving structure while reducing complexity

SRAI - Statistics, Reasoning and Artificial Intelligence

Book 1: Mathematical Foundations | Lesson 7

Production unit	PU-B01-C07
Version	v0.1
Canonical notebook	M1_N07_svd_pca_dimensionality_reduction.ipynb
Data boundary	Synthetic instructional data only
Core safeguard	Every dimensionality reduction is also an information-removal decision.

1. Why dimensionality reduction requires judgment

A national dashboard can contain dozens of correlated indicators. Compression can reveal dominant patterns and make analysis tractable, but it can also hide inequality, rare events or small dimensions that matter operationally. The analytical question is therefore not only how much variance is retained, but what information is removed and who could be affected.

Decision contract: Before reducing dimensions, record the decision purpose, variables, units, scaling rule, retained-dimension rule, reconstruction metric, subgroup checks and escalation owner.

2. Singular value decomposition

For any real matrix A with m rows and n columns, the singular value decomposition is:

$$A = U\Sigma V^T$$

- U contains orthonormal output directions (left singular vectors).
- Σ contains nonnegative singular values ordered from largest to smallest.
- V contains orthonormal input directions (right singular vectors).

Reconstruction must be verified numerically, together with $U^T U = I$ and $V^T V = I$. Singular-vector signs may flip without changing the represented subspace.

3. Low-rank approximation and information loss

Keeping only the first k singular directions gives the truncated approximation:

$$A_k = U_k \Sigma_k V_k^T$$

The Eckart-Young result makes A_k the best rank- k approximation under the Frobenius norm. The numerical reconstruction error is:

$$\|A - A_k\|_F$$

The fraction of singular-value energy associated with direction i is:

$$\sigma_i^2 / \sum_j \sigma_j^2$$

A small numerical error does not prove that the removed information is substantively unimportant. Rare harms, minority patterns and threshold events can occupy low-variance directions.

4. PCA from centered data

Let X_c be the column-centered data matrix. Applying SVD gives:

$$X_c = U\Sigma V^T \quad \text{and} \quad Z = U\Sigma$$

The columns of V are principal directions and Z contains component scores. PCA creates orthogonal directions ordered by explained variance. Centering is essential. Standardization is often appropriate when variables use different scales, but it changes the geometry and therefore the analytical question.

5. Interpretation and governance

- Treat loadings as mathematical associations, not causal effects.
- Do not name a component “development,” “resilience” or similar without domain evidence.
- Report retained variance and reconstruction loss, but also inspect subgroup and rare-event loss.
- Document whether data were centered, standardized or otherwise transformed.
- Keep a reproducible route back to the original variables and observations.

6. Worked example: national indicators

Suppose six regions are described by agricultural output, food availability, electricity access, health capacity and transport reliability. Standardize the five indicators, compute two components and inspect both explained variance and loadings. A high first-component variance share can summarize a common service-access pattern, but it does not authorize a causal label or the removal of indicators needed for equity monitoring.

Required preservation-and-loss statement: State what the retained components reproduce, which variables dominate them, which observations have the largest residuals, and what policy-relevant patterns could disappear.

7. AI applications and failure modes

- Applications: denoising, semantic analysis, recommender systems, embedding compression, visualization and low-rank adaptation.
- Failure mode: equating explained variance with universal importance.
- Failure mode: selecting k only from a convenient threshold without downstream validation.
- Failure mode: fitting preprocessing on the full dataset and leaking test information.
- Failure mode: interpreting arbitrary component signs as substantive direction.

8. Practical laboratory

Use the canonical notebook to verify SVD reconstruction and orthogonality, compare rank- k approximations, derive PCA from SVD, compare with scikit-learn and test compression on synthetic indicator data. Run all cells in sequence in a clean environment.

9. Key insight

SVD reveals dominant input-output directions. PCA compresses centered data while quantifying variation retained and lost. Responsible use requires an explicit record of what the compression preserves, what it removes and how those losses affect the intended decision.

10. References and further reading

- Golub, G. H., and Van Loan, C. F. Matrix Computations.
- Jolliffe, I. T., and Cadima, J. Principal component analysis: a review and recent developments.
- Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness.